

Mathematics, 2018 EC, entrance exam, version 6

Grade 12 · Mathematics · 2018 E.C.

Time allowed: 120 minutes

Questions: 51

Total marks: 51

Name _____ Grade _____ ID No. _____ Date _____

1. Which of the following is true about the relation between the set of integers ($\mathbb{Z}$), positive rational numbers ($\mathbb{Q}^+$) and rational numbers ($\mathbb{Q}$)? (1)
 - A. $\mathbb{Z} \subseteq \mathbb{Q}^+$
 - B. $\mathbb{Z} \cap \mathbb{Q}^+ = \mathbb{Q}^+$
 - C. $\mathbb{Q}^+ \cup \mathbb{Z} \subseteq \mathbb{Q}$
 - D. $\mathbb{Z} = \mathbb{Q}^+$
2. Which of the following is true? (1)
 - A. the set of irrational numbers are closed under multiplication
 - B. the sum of two irrational numbers is always irrational
 - C. the product of two irrational numbers is rational
 - D. the set of rational numbers is closed under addition and multiplication
3. If the system $\{x + by = 10, ax + 2y = 20\}$ has infinite solutions, then values of a and b respectively? (1)
 - A. 2 and 1
 - B. -1 and 2
 - C. 1 and 2
 - D. 1 and 1
4. The sum of two numbers is 75, two-third of larger number is 10 more than the smaller. What are the numbers? (1)
 - A. 30 and 45
 - B. 35 and 40
 - C. 24 and 51
 - D. 25 and 50
5. Which of the following is the solution set of $1 - 2|x+2| = 6$? (1)
 - A. $\{\}$
 - B. $\{1/2, 9/2\}$
 - C. $\{5, 9\}$
 - D. $\{-4\}$
6. The solution set of $|5 - 3x| + 9 > 17$ is: (1)
 - A. $(-\infty, -1] \cup [13/3, \infty)$
 - B. $(-\infty, 13/3) \cup (-1, \infty)$
 - C. $(-1, \infty)$
 - D. $(-\infty, -1) \cup (13/3, \infty)$
7. If $(5^x)^3 \cdot 8^x = 10\sqrt{10}$, then x is: (1)
 - A. $1/4$
 - B. $1/2$
 - C. $\sqrt{6}/6$
 - D. $1/6$
8. The solution set of the equation $(3^x)^2 \cdot 4^x = 6\sqrt{6}$ is: (1)
 - A. $x=1$
 - B. $x=0.75$
 - C. $x=2$
 - D. $x=\sqrt{2}$
9. If $(16)^x = 1/8$, then x is: (1)
 - A. $-4/3$
 - B. $4/3$
 - C. $3/4$
 - D. $-3/4$

10. Which of the following interval may not contain a zero of $f(x)=x^4-7x^2+10$? (1)
- A. $(-2,-1)$ B. $(1,2)$
C. $(-1,0)$ D. $(2,3)$
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11. The solution set of the equation $(3)(2^{4x})(5^{2x}) = 1200$ is: (1)
- A. 2 B. 4
C. 1 D. 10
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12. The solution set of the inequality $2x^2 - 5x - 7 > 0$ is: (1)
- A. $(1, 7/2)$ B. $(-\infty, 7) \cup (2, \infty)$
C. $(-\infty, -1) \cup (7/2, \infty)$ D. $(-7, 2)$
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13. The solution set of the inequality $-x^2 + 7x - 12 \leq 0$ is: (1)
- A. $(-\infty, \infty)$ B. $[3, 4]$
C. $(-\infty, 3] \cup [4, \infty)$ D. $\emptyset$
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14. The solution set of the inequality $x^2 - 9 \leq 0$ is: (1)
- A. $(-\infty, -3] \cup [3, \infty)$ B. $[-3, 3]$
C. $(-\infty, -9] \cup [9, \infty)$ D. $[-9, 9]$
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15. In the quadratic equation $(k-1)x^2 - 4x + 2k = 0$, if the sum of its roots is 2, then $k =$? (1)
- A. -4 B. 3
C. -3 D. 4
-
16. If 'a' and 'b' are the roots of $x^2 + 3x + 2 = 0$, then $a^2 + b^2 =$? (1)
- A. 13 B. 10
C. 5 D. -13
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17. What is the range of the inverse of $R = \{(x,y): y = \sqrt{1 - |x+2|}\}$? (1)
- A. $\{y: y \geq -2\}$ B. $\{y: -3 \leq y \leq -1\}$
C. $\{y: y \leq -2\}$ D. $\{y: 1 \leq y \leq 3\}$
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18. Which one of the following is true about Signum, absolute value and greatest integer functions? (1)
- A. $\text{sgn}(x) = \pm|x|, \forall x \in \mathbb{R}$ B. $\text{sgn}(x) \leq |x|, \forall x \geq 0$
C. $\text{sgn}(x) \leq |x|, \forall x \in \mathbb{R}$ D. $|x| = x \cdot \text{sgn}(x), \forall x \in \mathbb{R}$
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19. Let $f(x)=3x-4$ and $g(x)=mx+b$. For what m, b is $(f \circ g)(x)=x=(g \circ f)(x)$? (1)
- A. $m=1/3, b=4/3$ B. $m=-3, b=4$
C. $m=1/3, b=4$ D. $m=1/3, b=3/4$
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20. What is the partial fraction form of $(3x^2-2x-4)/(x^3+2x)$? (1)
- A. $2/x + (5x+2)/(x^2+2)$ B. $-2/x + (5x+3)/(x^2+2)$
C. $-2/x + (5x-2)/(x^2+2)$ D. $2/x + (5x-2)/(x^2+2)$

21. Which one of the following is not true about $f(x) = (x - x^3) / [x^3(x - 1)]$? (1)

- A. Vertical asymptotes: $x=0$
- B. Horizontal asymptote: $y=0$
- C. Graph intersects horizontal asymptote at $(-1,0)$
- D. Graph intersects vertical line $x=0$

22. Equation of line through intersection of $x-2y+1=0$ and $2x+y-3=0$ and parallel to $3x-4y+2=0$? (1)

- A. $4x-3y-1=0$ B. $3x-4y+1=0$
C. $3x+4y+1=0$ D. $3x-4y-1=0$

23. A circle radius 1 is internally tangent to circle $x^2+y^2-2y=8$ at (3,1). Equation of smaller circle? (1)

- A. $x^2+(y-1)^2=1$ B. $(x-1)^2+(y-2)^2=1$
C. $(x-2)^2+(y-1)^2=1$ D. $(x-4)^2+(y-1)^2=1$

24. Class groups: A:15 students avg16; B:14 students avg17; C:11 students avg14. Overall average? (1)

- A. 15.3
B. 16.2
C. 15.8
D. 15.4

25. A box contains 4 red, 3 white, 5 blue balls. Three balls drawn without replacement. Probability all three are blue? (1)

- A. $\frac{1}{22}$ B. $\frac{1}{55}$
C. $\frac{1}{27}$ D. $\frac{5}{12}$

26. Variance of 20 observations is 16. If each observation is multiplied by $\frac{3}{2}$, then standard deviation of resulting observations? (1)

- A. 36
B. 24
C. 9
D. 6

27. In a group of 45 students, 25 play basketball, 30 play football, 19 play both. Probability that a student plays neither? (1)

- A. $\frac{4}{5}$ B. $\frac{2}{5}$
C. $\frac{1}{5}$ D. $\frac{2}{3}$

28. If $A = \begin{bmatrix} 0 & 4 & 1 \\ 2 & 1 & 5 \\ 0 & 3 & 2 \end{bmatrix}$ and $\det(2B) = 40$, then $\det(AB) = ?$ (1)

- A. 200
B. -200
C. 50
D. -50

29. If the inverse of $A = \begin{bmatrix} -1 & 0 & 1 \\ 0 & y & 0 \\ -1 & x & 1/2 \end{bmatrix}$ is $B = \begin{bmatrix} 1 & 0 & y \\ 0 & -1/2 & x \\ 2 & 0 & y \end{bmatrix}$, then x and y are? (1)

- A. 1 and -2
B. 0 and -2
C. 1 and 2
D. -2 and 1

30. Let $A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & -1 & 3 \\ -2 & 2 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}$. If $C = 3BA - 2A^T$, then (2,3) entry of C is? (1)

- A. 20
B. 18
C. 11
D. 14

31. If $A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} -3 & 2 \\ 2 & -1 \end{bmatrix}$, then which is not true? (1)

- A. $AB = BA$
B. $A+B$ is symmetric
C. $B-A$ is a scalar matrix
D. $A+B = A-B$

32. Which one is false about the sequence $\{1, -1/3, -5/3, -3, -13/3, -17/3, -7, \dots\}$? (1)
- A. It is AP with $d = -4/3$ B. nth term $a_n = (7-4n)/3$
 C. $a_n = (1/3)(7-4n)$ D. It is GP
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33. Which one is the nth term of the sequence $\{0.5, 0.55, 0.555, \dots\}$? (1)
- A. $5(1-1/10^n)$ B. $(1/9)(1-1/10^n)$
 C. $(5/9)(1-1/10^n)$ D. $(5/9)(5-1/10^n)$
-
34. The series $1 - 3/5 + 9/25 - 27/125 + \dots$ is equal to: (1)
- A. 1 B. $4/25$
 C. $5/8$ D. $5/2$
-
35. A boat worth Birr 125,000 loses 15% of its value each year. What worth after 5 years? (1)
- A. 65,250.78 B. 55,463.164
 C. 218,625.78 D. 251,419.648
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36. Let $\{a_n\}$ with $a_1 = -2$ and $a_n = 1/a_{n-1}$ for $n \geq 2$, then $a_{98} = ?$ (1)
- A. -2 B. $-1/2$
 C. $1/97$ D. $-1/4$
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37. Which one is not true about the sequence $a_n = \{1+3+5+\dots+(2n-1)\}/(3n^2-1)$? (1)
- A. $1/2 < a_n < 1/3$ B. $\text{glb} = 1/2$
 C. $\text{glb} = 1/3$ D. $\lim a_n = 1/3$
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38. Let $f(x) = (kx+3)/(2x-k)$. If $f'(1) = -1$, then $k = ?$ (1)
- A. 2 B. $-1/2$
 C. $1/2$ D. -5
-
39. Let $f(x) = (1/3)x - |x^2-4|$. The set of all critical numbers is: (1)
- A. $\{-2, 1/6, 2\}$ B. $\{-2, -1/6, 0, 1/6, 2\}$
 C. $\{-2, -1/6, 1/6, 2\}$ D. $\{-2, -1/6, 2\}$
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40. Which one is false about $f(x) = x^3 - 3x^2 + 3x + 1$? (1)
- A. f concave upward on $[1, \infty)$ B. f strictly increasing on $(-\infty, \infty)$
 C. f has local extreme at $x=1$ D. (1,2) is inflection point
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41. Maximum possible area of an isosceles triangle with perimeter 6 m? (1)
- A. 2 m^2 B. $\sqrt{3} \text{ m}^2$
 C. $2\sqrt{3} \text{ m}^2$ D. 3 m^2
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42. A ladder 6 m long slides. Bottom moves at $3/4 \text{ m/s}$. Find angle between top and wall when angle changes at $\sqrt{2}/8 \text{ rad/s}$. (1)
- A. $\pi/2$ B. $\pi/3$
 C. $\pi/6$ D. $\pi/4$
-
43. Which point on graph of $f(x) = 2x-1$ is closest to (1,2)? (1)
- A. $(8/5, -4/5)$ B. $(7/5, 9/5)$
 C. $(4/5, 3/5)$ D. $(8/5, 4/5)$

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44. Volume generated by revolving region between $f(x)=x^2$ and $g(x)=4x-x^2$ about x-axis? (1)
- A. $224\pi/3$ B. $22\pi/3$
C. $32\pi/3$ D. $24\pi/3$
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45. Angle between vectors $u = m \mathbf{i} + m\sqrt{3} \mathbf{j}$ and $v = m\sqrt{3} \mathbf{i} + m \mathbf{j}$? (1)
- A. 90° B. 60°
C. 30° D. 53°
-
46. Image of line $l: 3x+2y=6$ after rotation of 90° about point $P(2,3)$ is: (1)
- A. $2x-3y=6$ B. $3x-2y=1$
C. $3x-2y=-6$ D. $2x-3y=1$
-
47. If the image of point $P(-2,3)$ is $P'(6,7)$ under a line reflection, then the line of reflection is: (1)
- A. $2y = x + 9$ B. $y = -2x + 9$
C. $y = 2x - 9$ D. $y = -x + 9$
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48. In triangle ABC, $AB=7$, $BC=8$, $m\angle B=120^\circ$. Find AC and $\sin\angle A$? (1)
- A. $\sqrt{57}$ and $4\sqrt{3}/13$ B. $\sqrt{57}$ and $-4\sqrt{3}/13$
C. 13 and $23/26$ D. 13 and $4\sqrt{3}/13$
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49. Points $P(3,3,2)$ and $Q(2,0,4)$ on sphere with center on z-axis. Which point is outside? (1)
- A. $(-4,3,0)$ B. $(2,-2,1)$
C. $(3,1,1)$ D. $(0,0,0)$
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50. Equation of tangent to circle $x^2+y^2-2x-2y-23=0$ at $P(-3,4)$ is: (1)
- A. $3y+4x=24$ B. $3x-y=24$
C. $4x-3y=0$ D. $4x-3y+24=0$
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51. Let ℓ be line containing $(1,5)$ with angle of inclination $\theta=135^\circ$. Distance from ℓ to $(-1,1)$ is: (1)
- A. $2\sqrt{2}$ B. $3\sqrt{2}$
C. $4\sqrt{2}$ D. $6\sqrt{2}$
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— END OF EXAMINATION —